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STATISTICS

Session 4: Measurement Error

Measurement

- We often work with numbers as if they are straightforward facts. These numbers can be useful. They may even be the best information we have.

- But they are not the same thing as the thing we are trying to understand.
  - A wine score ≠ wine quality
  - An assignment score ≠ understanding
  - Speedometer reading ≠ Speed

- Each number is produced by a process. And every process can introduce uncertainty.



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Random Error

- **Not all errors are the same.** Imagine a bathroom scale. Suppose you step on it five times in a row and it gives you the following: 70.1, 69.9, 70.3, 69.9, 70.0

- The readings are not identical, but they are close. The differences might be due to how you stood on the scale, tiny changes in the floor, or small imperfections in the instrument. This is an example of random error.

- Random error is the kind of error that varies from measurement to measurement. It does not follow an obvious direction in a single observation, but over repeated measurements we may begin to see its pattern.



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Systematic Error

- Now imagine a different scale. This scale is faulty. Every time you step on it, it adds ~2 kilograms.
- If your true weight is 70 kilograms, the scale might show: 72.1, 71.8, 72.3, 71.9, 72.0
- Repeated measurement does not automatically fix systematic error. If a scale is always 2 kilograms too high, weighing yourself one hundred times will not solve the problem.



Random and Systematic Error

Observed value = True value + Systematic Error + Random Error

Type of Error	Description
Random	Measurement varies unpredictably from one observation to the next.
Systematic	Measurement is consistently pushed in one direction.

Joseph Hobson Jagger

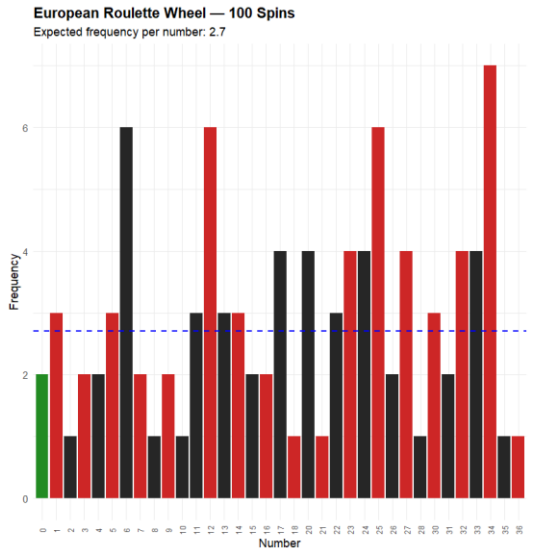


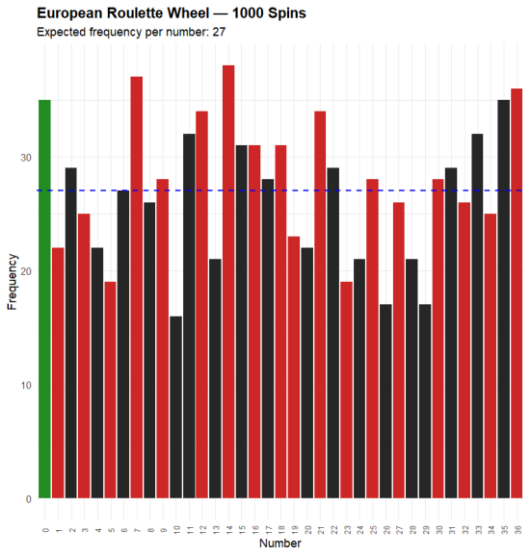
- Engineer and Mechanic
- Industry Businessman
- Not a mathematician / statistician but "a man who broke the bank at Monte Carlo" by recognising systematic error at the roulette table.

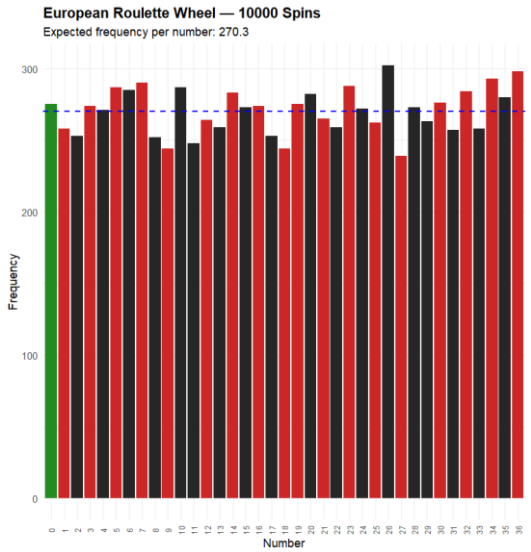
- A signal is a real pattern that tells us something about the process being measured. In Jagger's case, the signal was the wheel's physical bias.
- Noise is random variation that looks meaningful but does not reflect a real underlying pattern. A fair wheel might make one number appear usually often for a while, but that does not mean the number is truly favoured.

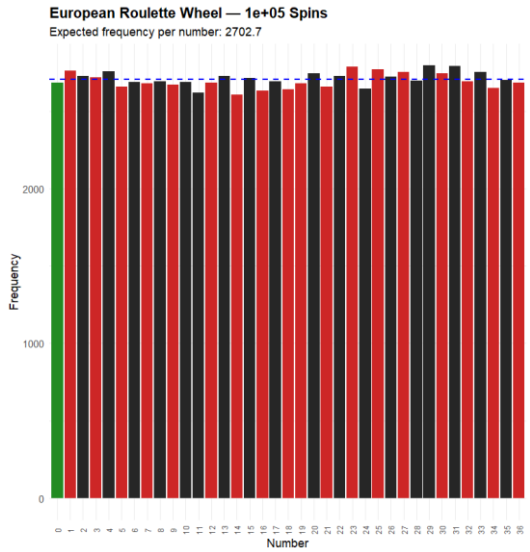
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Joseph Hobson Jagger









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Sequence

An ordered succession of elements, such as points or numbers.

Series

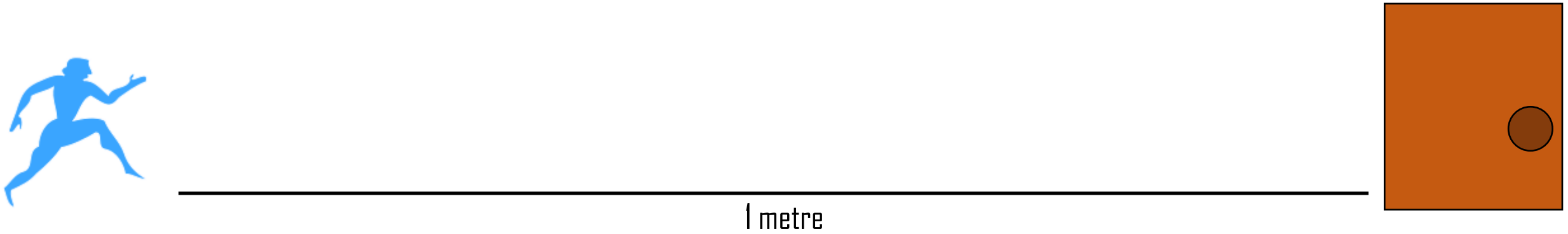
The sum of a sequence of numbers.

Limit

If the elements of a sequence seem to heading somewhere, such as an endpoint or a particular number, then that is the limit of the sequence.

Calculus & Statistics – Zeno's paradox

- Suppose you are standing at a particular point in a room and you want to reach the door (which happens to be 1m away). Now before you reach the door you must reach the ½ way point. But before you reach the ½ point, you need to reach ½ way to the ½ way point (i.e. ¼ way). And so on, and so on.
- Zeno argued that because the sequence goes on forever, you will have to travel an infinite number of finite distances. Zeno concluded that you must take an infinite amount of time.

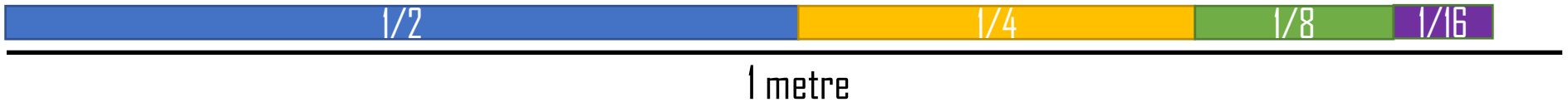


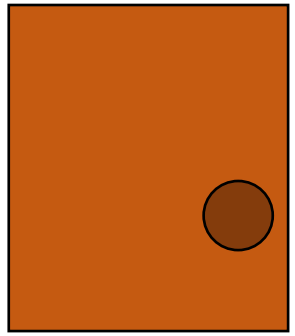
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Calculus & Statistics – Zeno's paradox

- The pattern looks like that if we include more intervals, the greater the sum of the distances we obtain. But Zeno is not correct when he says that the sum is headed towards infinity.
- Instead, the numbers seem to be approaching 1.
- A modern mathematician, using the language of calculus, would say: 1 metre is the limit of this sequence of distances.





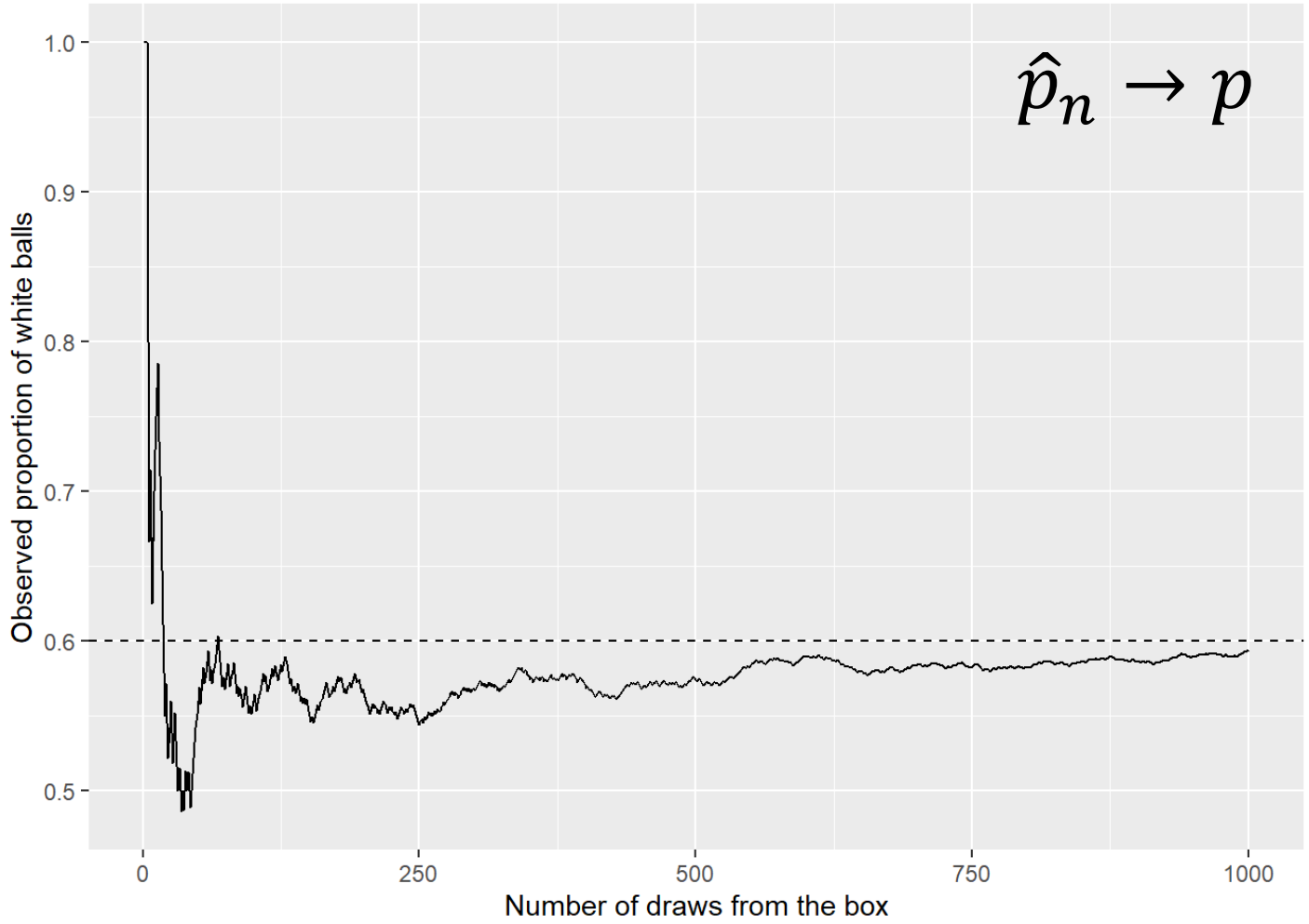


$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1$$

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Jacob Bernoulli & The Law of Large Numbers





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Jacob Bernoulli & The Law of Large Numbers

Repeated observations can reduce random error. They can help us see stable patterns. They can allow us to estimate probabilities that cannot be calculated in advance. But they do not remove the need to ask how the data were produced.

The Law of Small Numbers



Sherry Lansing

- Former studio executive, Universal Music Group
- Director of the board, Paramount Skydance
- Chairwoman and CEO, Paramount Pictures
- President of Production, 20th Century Fox

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The Law of Small Numbers

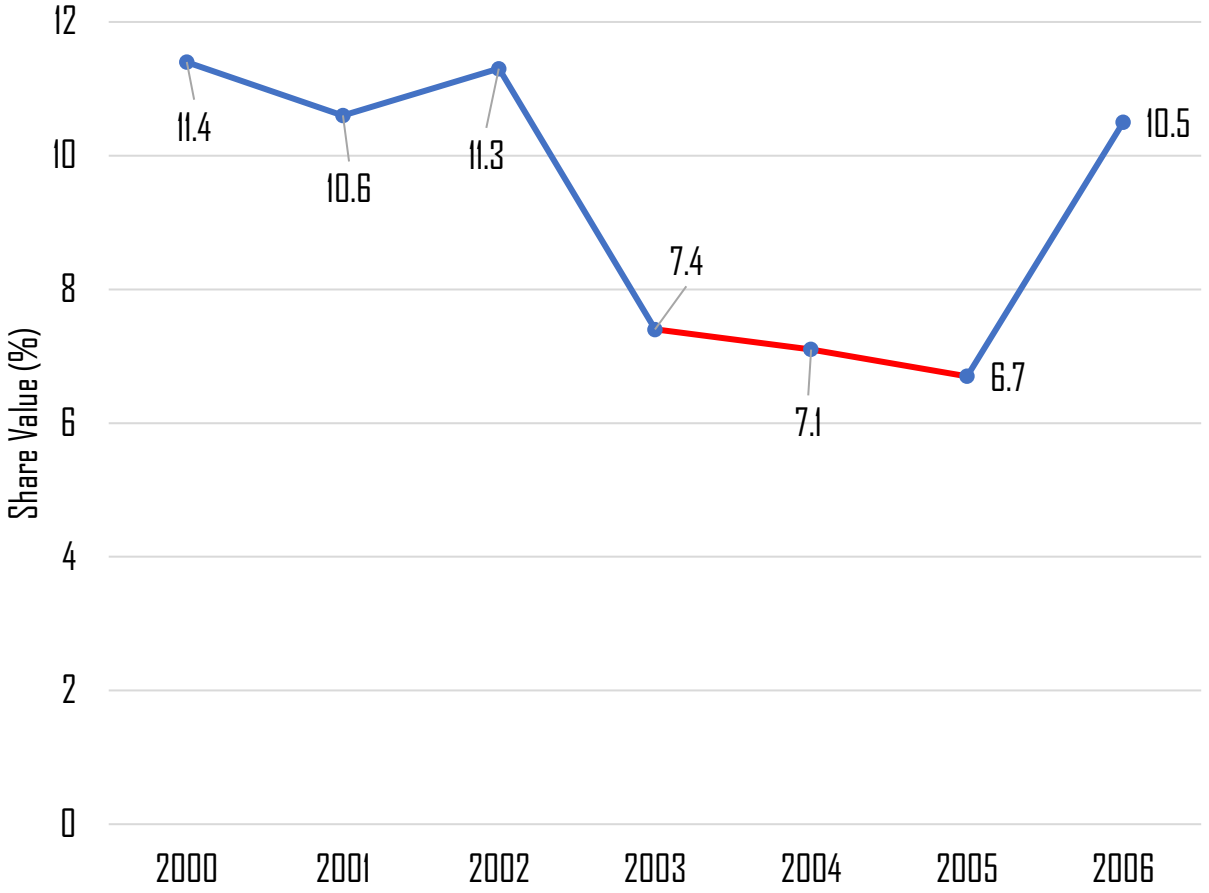


Lansing produces several award winning movies, Revenue increases.

Lansing produces underperforming movies. Revenue decreases.

Lansing produces some of Paramounts best performing movies

Lansing let go



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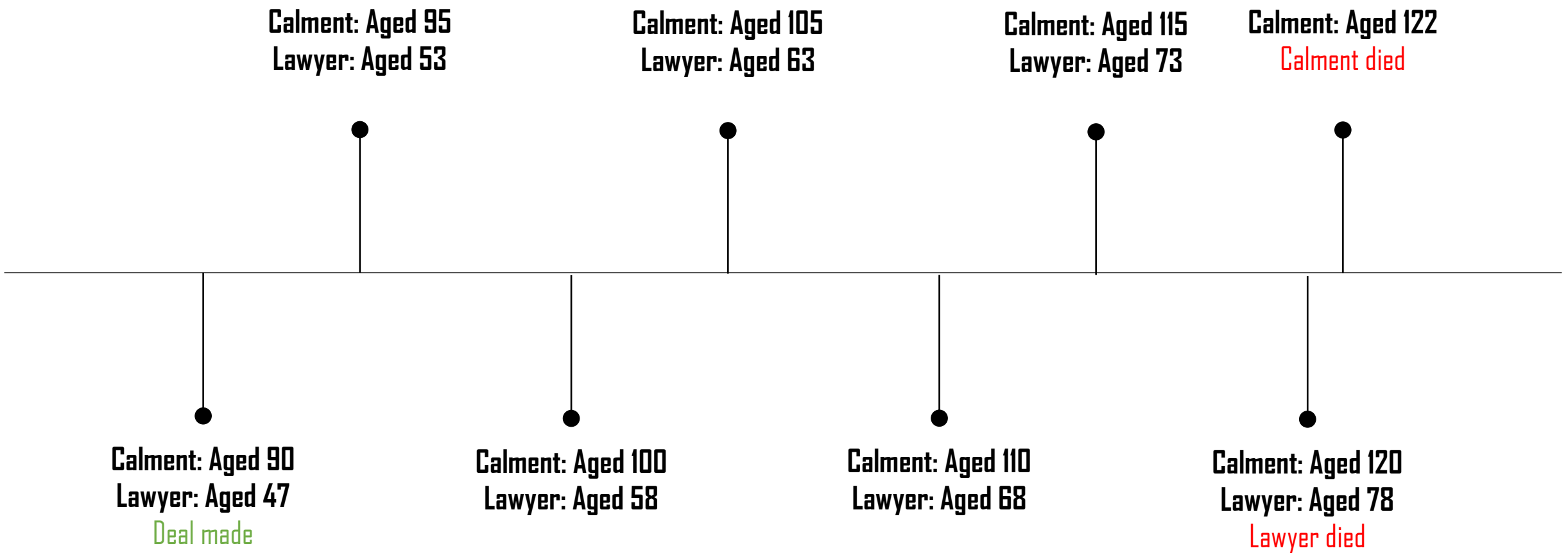
The Law of Small Numbers





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The Law of Small Numbers



The Law of Small Numbers

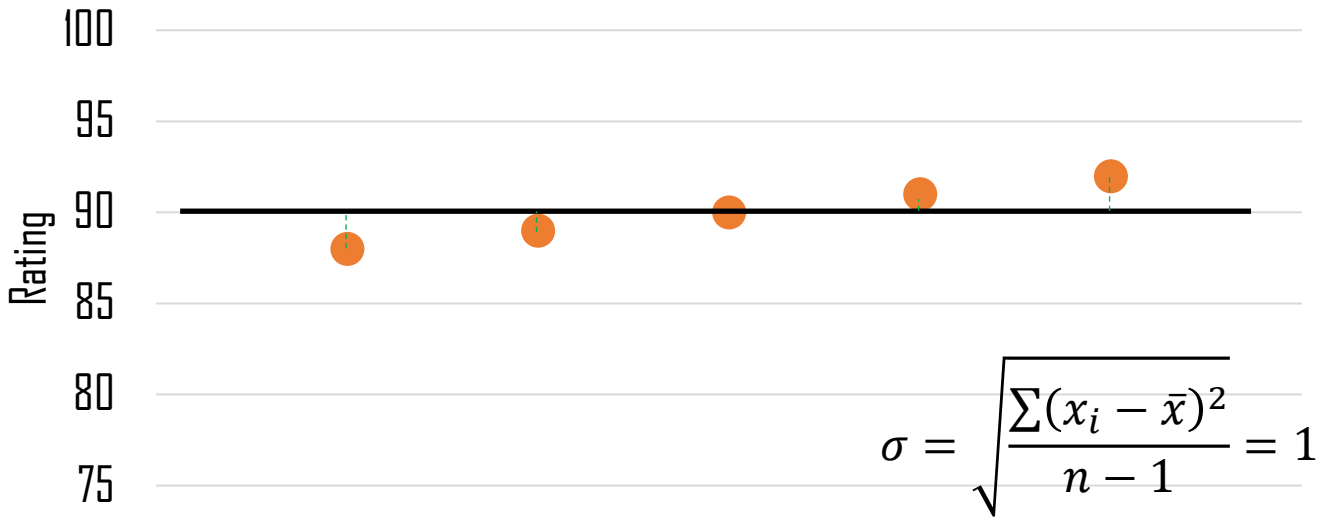
$$P(\text{live to } 100) \neq P(\text{live to } 100 \mid \text{already lived to } 90)$$

Probability statement	Plain-language meaning	Why this matters
P(live to 100)	The chance that a newborn eventually reaches 100.	Includes all early and middle-life risks.
P(live to 100 | already lived to 90)	The chance that a 90-year-old reaches 100.	Only includes people who have already survived to age 90.

The Shape of Chance



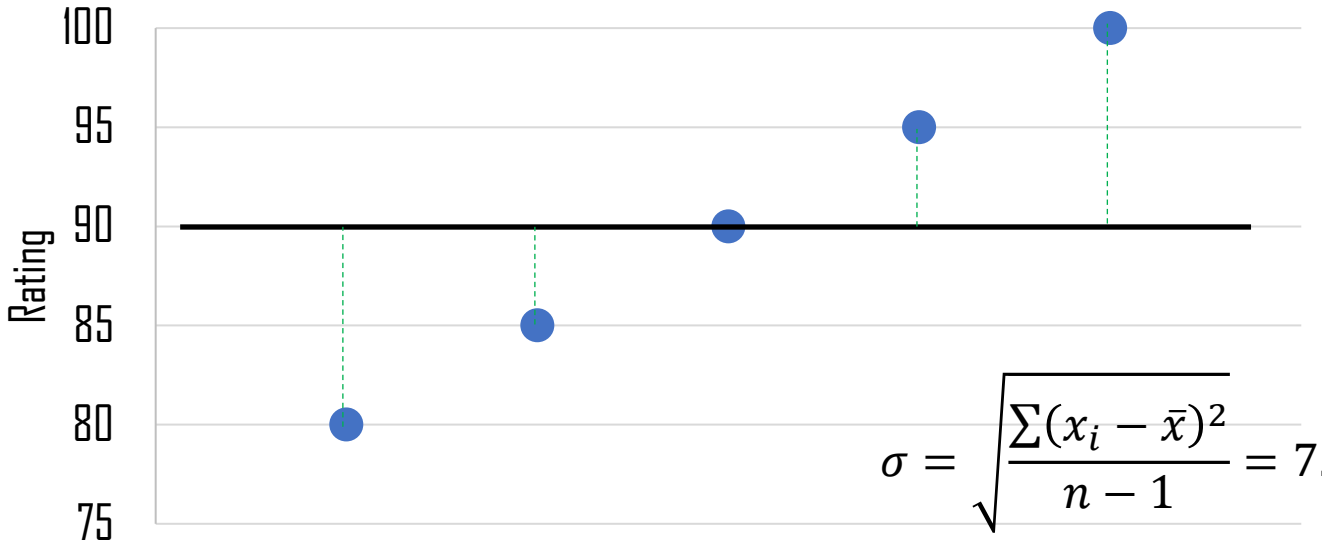
Critic	x	$\bar{x}$	$x - \bar{x}$	$(x - \bar{x})^2$
Critic 1	88	90	-2	4
Critic 2	89	90	-1	1
Critic 3	90	90	0	0
Critic 4	91	90	1	1
Critic 5	92	90	2	4



$$\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 1.58$$



Critic	x	$\bar{x}$	$x - \bar{x}$	$(x - \bar{x})^2$
Critic 1	80	90	-10	100
Critic 2	85	90	-5	25
Critic 3	90	90	0	0
Critic 4	95	90	5	25
Critic 5	100	90	10	100

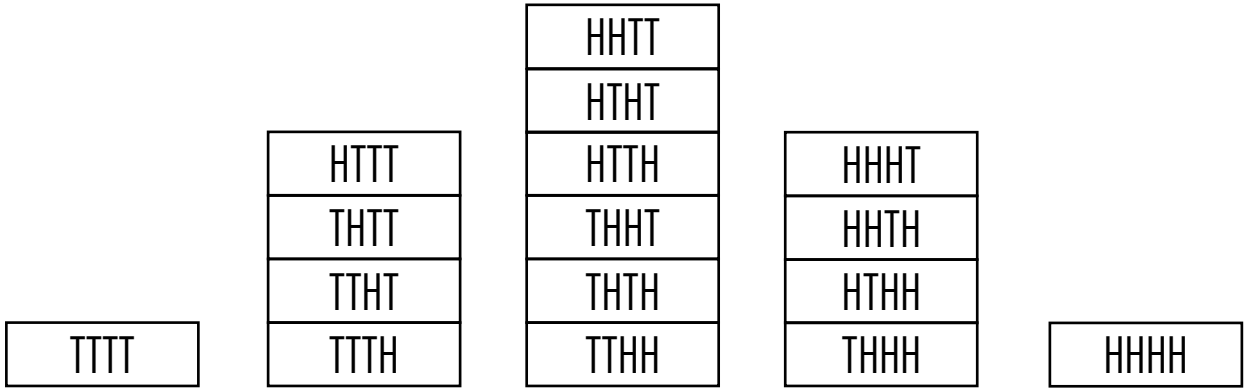


$$\sigma = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 7.91$$

The Shape of Chance

Context	What the average tells us	What variation tells us
Wine ratings	The centre of critics' ratings	How much critics agree or disagree
Exam marks	The centre of student performance	How spread out student performance is
Income	The centre of household income	How unequal incomes are
Life expectancy	The centre of lifespans	How much individual lifespans differ
Polls	The estimated level of support	How uncertain the estimate may be

The Shape of Chance



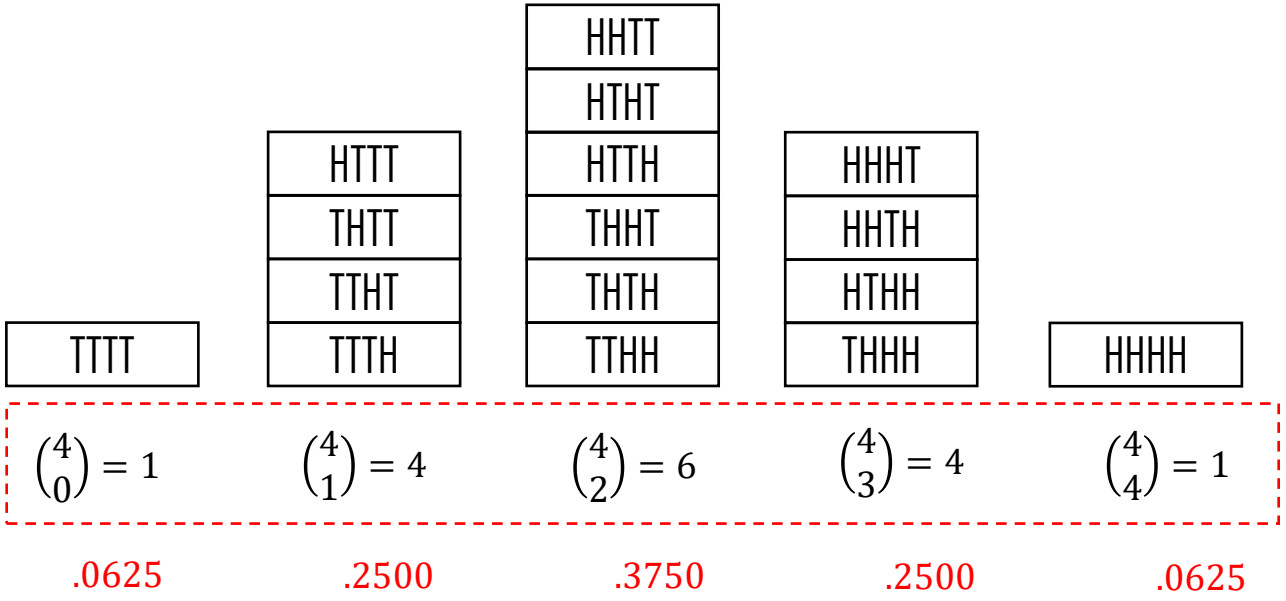
$\binom{4}{0} = 1$ $\binom{4}{1} = 4$ $\binom{4}{2} = 6$ $\binom{4}{3} = 4$ $\binom{4}{4} = 1$

.0625 .2500 .3750 .2500 .0625

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}
$$

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The Shape of Chance



$$P(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P(0 \text{ heads}) = \binom{4}{0} 0.5^0 (1-0.5)^{4-0} = .0625$$

$$P(1 \text{ heads}) = \binom{4}{1} 0.5^1 (1-0.5)^{4-1} = .2500$$

$$P(2 \text{ heads}) = \binom{4}{2} 0.5^2 (1-0.5)^{4-2} = .3750$$

$$P(3 \text{ heads}) = \binom{4}{3} 0.5^3 (1-0.5)^{4-3} = .2500$$

$$P(4 \text{ heads}) = \binom{4}{4} 0.5^4 (1-0.5)^{4-4} = .2500$$

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